

KROPYVNYSTKYI, Ukraine

WMO: 337110

Lat: **48.5436N**Lon: **32.2833E**Elev: **561**StdP: **14.40**Time Zone: **2.00 (EUE)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.0	5.1	-6.5	4.0	0.1	-0.3	5.5	6.0	23.5	22.5	21.5	24.2	6.2	330	0.560

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.3	90.6	67.5	87.3	66.7	84.3	65.8	70.7	83.4	69.3	81.6	67.9	79.8	9.8	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.7	100.5	75.1	65.3	95.6	73.7	63.9	91.0	72.5	34.9	83.6	33.7	81.7	32.5	79.7	75.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.9	18.8	16.9	-7.6	94.7	6.3	3.8	-12.1	97.4	-15.8	99.6	-19.3	101.7	-23.9	104.4		
(5)			-8.1	72.8	6.1	1.4	-12.5	73.7	-16.1	74.5	-19.5	75.3	-24.0	76.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.3	24.6	27.1	36.2	49.9	60.3	66.7	70.7	69.4	59.6	47.8	37.5	28.6	
	DBStd	18.24	10.13	10.23	8.37	7.31	7.47	6.13	5.41	6.26	7.20	8.03	8.64	9.66	
	HDD50	3152	786	641	431	89	10	0	0	0	9	139	384	663	
	HDD65	6613	1251	1061	894	454	183	53	12	25	190	536	826	1128	
	CDD50	2537	0	0	3	86	330	502	641	601	297	70	7	0	
	CDD65	522	0	0	0	1	38	105	188	161	28	1	0	0	
	CDH74	5446	0	0	0	26	430	1035	1842	1776	327	10	0	0	
	CDH80	1830	0	0	0	4	99	293	642	707	85	0	0	0	
Wind	WSAvg	8.5	9.1	9.5	9.9	9.2	8.1	7.7	7.2	7.4	7.7	7.9	9.0	9.0	
Precipitation	PrecAvg	19.6	1.3	1.1	1.1	1.4	1.8	2.6	2.5	1.9	1.5	1.1	1.5	1.6	
	PrecMax	29.6	5.1	2.7	2.6	3.1	4.0	8.5	5.9	5.2	7.0	4.2	6.1	3.8	
	PrecMin	12.9	0.3	0.0	0.1	0.1	0.1	0.4	0.0	0.0	0.0	0.0	0.2	0.2	
	PrecStd	3.9	0.9	0.6	0.7	0.7	0.7	1.7	1.4	1.2	1.4	1.0	1.2	0.9	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.5	49.7	63.0	77.0	87.8	90.0	93.7	96.2	87.5	74.6	61.5	50.6	
		MCWB	43.4	44.5	50.0	56.8	63.7	67.3	69.1	68.0	63.5	59.3	54.9	48.5	
	2%	DB	41.7	45.0	57.3	71.2	82.4	86.4	89.9	91.7	82.1	69.1	55.7	47.0	
		MCWB	39.7	41.9	46.9	54.7	62.4	66.3	68.2	67.4	62.0	57.6	52.1	45.1	
	5%	DB	38.6	41.8	52.3	67.4	79.0	83.5	86.7	87.5	77.5	63.9	52.4	43.8	
		MCWB	37.2	39.4	44.3	52.7	61.6	65.7	67.8	66.5	60.4	55.8	50.1	42.5	
	10%	DB	36.4	38.8	48.2	63.5	74.9	80.4	83.4	83.6	73.2	59.7	49.2	40.7	
		MCWB	35.2	36.7	42.0	51.0	60.1	64.9	67.0	65.3	59.1	53.7	47.2	39.5	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.8	45.7	51.1	59.6	67.1	71.4	72.6	72.0	67.3	63.3	56.1	49.2
			MCDB	45.2	48.5	61.2	74.8	80.0	82.9	85.0	86.3	81.0	70.4	59.8	50.3
2%		WB	39.9	42.6	47.6	56.5	64.8	69.5	71.0	70.1	64.3	59.6	53.0	45.3	
		MCDB	41.1	44.5	55.8	67.3	77.3	80.9	83.4	83.9	76.2	64.8	55.6	46.4	
5%		WB	37.4	39.5	45.0	54.3	63.0	67.9	69.7	68.3	62.3	56.9	50.3	42.6	
		MCDB	38.4	41.6	50.8	63.7	75.2	79.0	82.1	81.9	72.8	62.6	52.0	43.8	
10%		WB	35.4	37.0	42.6	52.5	61.2	66.2	68.4	66.8	60.6	54.4	47.6	39.6	
		MCDB	36.3	38.6	47.3	61.1	72.4	76.4	79.9	79.6	70.4	59.3	49.0	40.5	
Mean Daily Temperature Range		5% DB	MDBR	8.7	10.1	13.4	18.4	20.1	19.3	20.3	21.5	18.8	15.1	9.8	8.5
			MCDBR	8.4	12.5	21.3	24.7	25.3	24.2	24.8	27.3	26.3	21.6	14.2	10.2
	MCWBR		7.2	9.8	12.7	11.5	10.2	9.0	8.2	8.7	10.6	11.6	10.5	8.7	
	5% WB	MCDBR	7.7	11.8	19.0	21.0	21.8	20.2	21.0	22.8	21.5	18.1	12.3	9.6	
		MCWBR	7.0	9.6	12.3	10.4	9.8	8.8	8.0	8.3	10.3	11.3	10.0	8.5	
Clear-Sky Solar Irradiance	taub		0.321	0.352	0.383	0.442	0.414	0.406	0.437	0.432	0.404	0.367	0.349	0.324	
	taud		2.231	2.182	2.215	2.101	2.267	2.304	2.223	2.229	2.278	2.365	2.327	2.303	
	Ebn at Noon		227	247	260	255	269	272	261	256	251	242	217	211	
	Edh at Noon		25	33	37	47	41	40	43	41	35	27	23	21	
All-Sky Solar Radiation	RadAvg		306	575	960	1435	1842	1978	1942	1742	1192	718	300	212	
	RadStd		59	109	151	133	153	149	107	104	168	101	50	44	